# Causality Assessment

# Naranjo Algorithm – Scoring Worksheet and Justification Report

The following assessment was completed using the information available in the case pack. The Naranjo 10-question scale was used for each case. Where the case did not provide enough information to answer a question, it was marked as Unknown and no score was added.

Naranjo interpretation: 9 or more = Definite; 5–8 = Probable; 1–4 = Possible; 0 or less = Doubtful.

## Case Summary

| Case | Suspect product | Reaction | Score | Result |
|---|---|---|---|---|
| SYN-PV-011 | Metformin | Nausea | 4 | Possible |
| SYN-PV-012 | Amlodipine | Peripheral oedema | 4 | Possible |
| SYN-PV-013 | Ciprofloxacin | Tendon pain | 4 | Possible |
| SYN-PV-014 | Sertraline | Dizziness | 1 | Possible |
| SYN-PV-015 | Omeprazole | Headache | 4 | Possible |
| SYN-PV-016 | Losartan | Fatigue | 2 | Possible |
| SYN-PV-017 | Amoxicillin | Rash | 4 | Possible |
| SYN-PV-018 | Atorvastatin | Myalgia | 4 | Possible |
| SYN-PV-019 | Metformin | Nausea | 2 | Possible |
| SYN-PV-020 | Amlodipine | Peripheral oedema | 2 | Possible |
| SYN-PV-021 | Ciprofloxacin | Tendon pain | 3 | Possible |
| SYN-PV-022 | Sertraline | Dizziness | 4 | Possible |
| SYN-PV-023 | Omeprazole | Headache | 1 | Possible |
| SYN-PV-024 | Losartan | Fatigue | 5 | Probable |
| SYN-PV-025 | Amoxicillin | Rash | 4 | Possible |
| SYN-PV-026 | Atorvastatin | Myalgia | 4 | Possible |
| SYN-PV-027 | Metformin | Nausea | 4 | Possible |
| SYN-PV-028 | Amlodipine | Peripheral oedema | 1 | Possible |
| SYN-PV-029 | Ciprofloxacin | Tendon pain | 4 | Possible |
| SYN-PV-030 | Sertraline | Dizziness | 5 | Probable |

## SYN-PV-011 – Metformin / Nausea

**Patient:** 23 years, Unknown  **Dose:** 5 mg once daily  **Route:** Oral
**Indication:** Hypertension  **Onset:** Day 19
**Dechallenge:** Unknown  **Rechallenge:** Positive
**Concomitant medicines:** None reported

| Q | Naranjo question | Answer | Score |
|---|---|---|---|
| 1 | Previous conclusive reports? | Unknown | 0 |
| 2 | Event after drug administered? | Yes | 2 |
| 3 | Improved on withdrawal/antagonist? | Unknown | 0 |
| 4 | Reappeared on re-administration? | Positive | 2 |
| 5 | Alternative causes? | Unknown | 0 |
| 6 | Reappeared with placebo? | Unknown | 0 |
| 7 | Drug level in toxic range? | Unknown | 0 |
| 8 | Dose-response relationship? | Unknown | 0 |
| 9 | Similar previous reaction? | Unknown | 0 |
| 10 | Objective evidence? | Unknown | 0 |

**Total score: 4  Causality: Possible**

**Justification:** The adverse event occurred after administration of Metformin, so the temporal relationship supports a positive answer to Naranjo question 2 (+2). Dechallenge information was not available, so question 3 was scored 0. A positive rechallenge was reported and therefore added +2 for question 4. The remaining questions were marked Unknown because the supplied case information does not provide sufficient evidence to score them. No assumption was made from the presence of concomitant medicines.

**Conclusion:** The final Naranjo score is in the Possible range. The timing of the event supports a possible relationship, but the available information is not sufficient to establish a stronger causal association.

## SYN-PV-012 – Amlodipine / Peripheral oedema

**Patient:** 24 years, Female  **Dose:** 5 mg once daily  **Route:** Oral
**Indication:** Hypertension  **Onset:** Day 4
**Dechallenge:** Unknown  **Rechallenge:** Positive
**Concomitant medicines:** None reported

| Q | Naranjo question | Answer | Score |
|---|---|---|---|
| 1 | Previous conclusive reports? | Unknown | 0 |
| 2 | Event after drug administered? | Yes | 2 |
| 3 | Improved on withdrawal/antagonist? | Unknown | 0 |
| 4 | Reappeared on re-administration? | Positive | 2 |
| 5 | Alternative causes? | Unknown | 0 |
| 6 | Reappeared with placebo? | Unknown | 0 |
| 7 | Drug level in toxic range? | Unknown | 0 |
| 8 | Dose-response relationship? | Unknown | 0 |
| 9 | Similar previous reaction? | Unknown | 0 |
| 10 | Objective evidence? | Unknown | 0 |

**Total score: 4  Causality: Possible**

**Justification:** The adverse event occurred after administration of Amlodipine, so the temporal relationship supports a positive answer to Naranjo question 2 (+2). Dechallenge information was not available, so question 3 was scored 0. A positive rechallenge was reported and therefore added +2 for question 4. The remaining questions were marked Unknown because the supplied case information does not provide sufficient evidence to score them. No assumption was made from the presence of concomitant medicines.

**Conclusion:** The final Naranjo score is in the Possible range. The timing of the event supports a possible relationship, but the available information is not sufficient to establish a stronger causal association.

# SYN-PV-013 – Ciprofloxacin / Tendon pain

**Patient:** 75 years, Female  **Dose:** 500 mg twice daily  **Route:** Oral
**Indication:** Diabetes  **Onset:** Day 5
**Dechallenge:** Negative  **Rechallenge:** Positive
**Concomitant medicines:** Vitamin D

| Q | Naranjo question | Answer | Score |
|---|---|---|---|
| 1 | Previous conclusive reports? | Unknown | 0 |
| 2 | Event after drug administered? | Yes | 2 |
| 3 | Improved on withdrawal/antagonist? | Negative | 0 |
| 4 | Reappeared on re-administration? | Positive | 2 |
| 5 | Alternative causes? | Unknown | 0 |
| 6 | Reappeared with placebo? | Unknown | 0 |
| 7 | Drug level in toxic range? | Unknown | 0 |
| 8 | Dose-response relationship? | Unknown | 0 |
| 9 | Similar previous reaction? | Unknown | 0 |
| 10 | Objective evidence? | Unknown | 0 |

**Total score: 4  Causality: Possible**

**Justification:** The adverse event occurred after administration of Ciprofloxacin, so the temporal relationship supports a positive answer to Naranjo question 2 (+2). The recorded dechallenge was negative, so no point was assigned for question 3. A positive rechallenge was reported and therefore added +2 for question 4. The remaining questions were marked Unknown because the supplied case information does not provide sufficient evidence to score them. No assumption was made from the presence of concomitant medicines.

**Conclusion:** The final Naranjo score is in the Possible range. The timing of the event supports a possible relationship, but the available information is not sufficient to establish a stronger causal association.

# SYN-PV-014 – Sertraline / Dizziness

**Patient:** 75 years, Female  **Dose:** 5 mg once daily  **Route:** Oral
**Indication:** Diabetes  **Onset:** Day 9
**Dechallenge:** Negative  **Rechallenge:** Negative
**Concomitant medicines:** Paracetamol

| Q | Naranjo question | Answer | Score |
|---|---|---|---|
| 1 | Previous conclusive reports? | Unknown | 0 |
| 2 | Event after drug administered? | Yes | 2 |
| 3 | Improved on withdrawal/antagonist? | Negative | 0 |
| 4 | Reappeared on re-administration? | Negative | -1 |
| 5 | Alternative causes? | Unknown | 0 |
| 6 | Reappeared with placebo? | Unknown | 0 |
| 7 | Drug level in toxic range? | Unknown | 0 |
| 8 | Dose-response relationship? | Unknown | 0 |
| 9 | Similar previous reaction? | Unknown | 0 |
| 10 | Objective evidence? | Unknown | 0 |

**Total score: 1  Causality: Possible**

**Justification:** The adverse event occurred after administration of Sertraline, so the temporal relationship supports a positive answer to Naranjo question 2 (+2). The recorded dechallenge was negative, so no point was assigned for question 3. A negative rechallenge was reported and therefore gave -1 for question 4. The remaining questions were marked Unknown because the supplied case information does not provide sufficient evidence to score them. No assumption was made from the presence of concomitant medicines.

**Conclusion:** The final Naranjo score is in the Possible range. The timing of the event supports a possible relationship, but the available information is not sufficient to establish a stronger causal association.

## SYN-PV-015 – Omeprazole / Headache

**Patient:** 33 years, Male  **Dose:** 20 mg once daily  **Route:** Oral
**Indication:** Hypertension  **Onset:** Day 16
**Dechallenge:** Unknown  **Rechallenge:** Positive
**Concomitant medicines:** None reported

| Q | Naranjo question | Answer | Score |
|---|---|---|---|
| 1 | Previous conclusive reports? | Unknown | 0 |
| 2 | Event after drug administered? | Yes | 2 |
| 3 | Improved on withdrawal/antagonist? | Unknown | 0 |
| 4 | Reappeared on re-administration? | Positive | 2 |
| 5 | Alternative causes? | Unknown | 0 |
| 6 | Reappeared with placebo? | Unknown | 0 |
| 7 | Drug level in toxic range? | Unknown | 0 |
| 8 | Dose-response relationship? | Unknown | 0 |
| 9 | Similar previous reaction? | Unknown | 0 |
| 10 | Objective evidence? | Unknown | 0 |

**Total score: 4  Causality: Possible**

**Justification:** The adverse event occurred after administration of Omeprazole, so the temporal relationship supports a positive answer to Naranjo question 2 (+2). Dechallenge information was not available, so question 3 was scored 0. A positive rechallenge was reported and therefore added +2 for question 4. The remaining questions were marked Unknown because the supplied case information does not provide sufficient evidence to score them. No assumption was made from the presence of concomitant medicines.

**Conclusion:** The final Naranjo score is in the Possible range. The timing of the event supports a possible relationship, but the available information is not sufficient to establish a stronger causal association.

## SYN-PV-016 – Losartan / Fatigue

**Patient:** 66 years, Unknown  **Dose:** 20 mg once daily  **Route:** Oral
**Indication:** Diabetes  **Onset:** Day 10
**Dechallenge:** Positive  **Rechallenge:** Negative
**Concomitant medicines:** None reported

| Q | Naranjo question | Answer | Score |
|---|---|---|---|
| 1 | Previous conclusive reports? | Unknown | 0 |
| 2 | Event after drug administered? | Yes | 2 |
| 3 | Improved on withdrawal/antagonist? | Positive | 1 |
| 4 | Reappeared on re-administration? | Negative | -1 |
| 5 | Alternative causes? | Unknown | 0 |
| 6 | Reappeared with placebo? | Unknown | 0 |
| 7 | Drug level in toxic range? | Unknown | 0 |
| 8 | Dose-response relationship? | Unknown | 0 |
| 9 | Similar previous reaction? | Unknown | 0 |
| 10 | Objective evidence? | Unknown | 0 |

**Total score: 2  Causality: Possible**

**Justification:** The adverse event occurred after administration of Losartan, so the temporal relationship supports a positive answer to Naranjo question 2 (+2). The event improved after withdrawal/dechallenge, supporting question 3 (+1). A negative rechallenge was reported and therefore gave -1 for question 4. The remaining questions were marked Unknown because the supplied case information does not provide sufficient evidence to score them. No assumption was made from the presence of concomitant medicines.

**Conclusion:** The final Naranjo score is in the Possible range. The timing of the event supports a possible relationship, but the available information is not sufficient to establish a stronger causal association.

# SYN-PV-017 – Amoxicillin / Rash

**Patient:** 57 years, Unknown  **Dose:** 5 mg once daily  **Route:** Oral
**Indication:** Hypertension  **Onset:** Day 21
**Dechallenge:** Unknown  **Rechallenge:** Positive
**Concomitant medicines:** Vitamin D

| Q | Naranjo question | Answer | Score |
|---|---|---|---|
| 1 | Previous conclusive reports? | Unknown | 0 |
| 2 | Event after drug administered? | Yes | 2 |
| 3 | Improved on withdrawal/antagonist? | Unknown | 0 |
| 4 | Reappeared on re-administration? | Positive | 2 |
| 5 | Alternative causes? | Unknown | 0 |
| 6 | Reappeared with placebo? | Unknown | 0 |
| 7 | Drug level in toxic range? | Unknown | 0 |
| 8 | Dose-response relationship? | Unknown | 0 |
| 9 | Similar previous reaction? | Unknown | 0 |
| 10 | Objective evidence? | Unknown | 0 |

**Total score: 4  Causality: Possible**

**Justification:** The adverse event occurred after administration of Amoxicillin, so the temporal relationship supports a positive answer to Naranjo question 2 (+2). Dechallenge information was not available, so question 3 was scored 0. A positive rechallenge was reported and therefore added +2 for question 4. The remaining questions were marked Unknown because the supplied case information does not provide sufficient evidence to score them. No assumption was made from the presence of concomitant medicines.

**Conclusion:** The final Naranjo score is in the Possible range. The timing of the event supports a possible relationship, but the available information is not sufficient to establish a stronger causal association.

## SYN-PV-018 – Atorvastatin / Myalgia

**Patient:** 56 years, Female  **Dose:** 500 mg twice daily  **Route:** Oral
**Indication:** Anxiety  **Onset:** Day 17
**Dechallenge:** Unknown  **Rechallenge:** Positive
**Concomitant medicines:** None reported

| Q | Naranjo question | Answer | Score |
|---|---|---|---|
| 1 | Previous conclusive reports? | Unknown | 0 |
| 2 | Event after drug administered? | Yes | 2 |
| 3 | Improved on withdrawal/antagonist? | Unknown | 0 |
| 4 | Reappeared on re-administration? | Positive | 2 |
| 5 | Alternative causes? | Unknown | 0 |
| 6 | Reappeared with placebo? | Unknown | 0 |
| 7 | Drug level in toxic range? | Unknown | 0 |
| 8 | Dose-response relationship? | Unknown | 0 |
| 9 | Similar previous reaction? | Unknown | 0 |
| 10 | Objective evidence? | Unknown | 0 |

**Total score: 4  Causality: Possible**

**Justification:** The adverse event occurred after administration of Atorvastatin, so the temporal relationship supports a positive answer to Naranjo question 2 (+2). Dechallenge information was not available, so question 3 was scored 0. A positive rechallenge was reported and therefore added +2 for question 4. The remaining questions were marked Unknown because the supplied case information does not provide sufficient evidence to score them. No assumption was made from the presence of concomitant medicines.

**Conclusion:** The final Naranjo score is in the Possible range. The timing of the event supports a possible relationship, but the available information is not sufficient to establish a stronger causal association.

## SYN-PV-019 – Metformin / Nausea

**Patient:** 29 years, Unknown  **Dose:** 5 mg once daily  **Route:** Oral
**Indication:** Anxiety  **Onset:** Day 9
**Dechallenge:** Positive  **Rechallenge:** Negative
**Concomitant medicines:** None reported

| Q | Naranjo question | Answer | Score |
|---|---|---|---|
| 1 | Previous conclusive reports? | Unknown | 0 |
| 2 | Event after drug administered? | Yes | 2 |
| 3 | Improved on withdrawal/antagonist? | Positive | 1 |
| 4 | Reappeared on re-administration? | Negative | -1 |
| 5 | Alternative causes? | Unknown | 0 |
| 6 | Reappeared with placebo? | Unknown | 0 |
| 7 | Drug level in toxic range? | Unknown | 0 |
| 8 | Dose-response relationship? | Unknown | 0 |
| 9 | Similar previous reaction? | Unknown | 0 |
| 10 | Objective evidence? | Unknown | 0 |

**Total score: 2  Causality: Possible**

**Justification:** The adverse event occurred after administration of Metformin, so the temporal relationship supports a positive answer to Naranjo question 2 (+2). The event improved after withdrawal/dechallenge, supporting question 3 (+1). A negative rechallenge was reported and therefore gave -1 for question 4. The remaining questions were marked Unknown because the supplied case information does not provide sufficient evidence to score them. No assumption was made from the presence of concomitant medicines.

**Conclusion:** The final Naranjo score is in the Possible range. The timing of the event supports a possible relationship, but the available information is not sufficient to establish a stronger causal association.

## SYN-PV-020 – Amlodipine / Peripheral oedema

**Patient:** 56 years, Unknown  **Dose:** 5 mg once daily  **Route:** Oral
**Indication:** Diabetes  **Onset:** Day 7
**Dechallenge:** Unknown  **Rechallenge:** Not done
**Concomitant medicines:** Paracetamol

| Q | Naranjo question | Answer | Score |
|---|---|---|---|
| 1 | Previous conclusive reports? | Unknown | 0 |
| 2 | Event after drug administered? | Yes | 2 |
| 3 | Improved on withdrawal/antagonist? | Unknown | 0 |
| 4 | Reappeared on re-administration? | Not done | 0 |
| 5 | Alternative causes? | Unknown | 0 |
| 6 | Reappeared with placebo? | Unknown | 0 |
| 7 | Drug level in toxic range? | Unknown | 0 |
| 8 | Dose-response relationship? | Unknown | 0 |
| 9 | Similar previous reaction? | Unknown | 0 |
| 10 | Objective evidence? | Unknown | 0 |

**Total score: 2  Causality: Possible**

**Justification:** The adverse event occurred after administration of Amlodipine, so the temporal relationship supports a positive answer to Naranjo question 2 (+2). Dechallenge information was not available, so question 3 was scored 0. Rechallenge was not done, so question 4 was scored 0. The remaining questions were marked Unknown because the supplied case information does not provide sufficient evidence to score them. No assumption was made from the presence of concomitant medicines.

**Conclusion:** The final Naranjo score is in the Possible range. The timing of the event supports a possible relationship, but the available information is not sufficient to establish a stronger causal association.

# SYN-PV-021 – Ciprofloxacin / Tendon pain

**Patient:** 33 years, Unknown  **Dose:** 5 mg once daily  **Route:** Oral
**Indication:** Infection  **Onset:** Day 9
**Dechallenge:** Positive  **Rechallenge:** Not done
**Concomitant medicines:** Paracetamol

| Q | Naranjo question | Answer | Score |
|---|---|---|---|
| 1 | Previous conclusive reports? | Unknown | 0 |
| 2 | Event after drug administered? | Yes | 2 |
| 3 | Improved on withdrawal/antagonist? | Positive | 1 |
| 4 | Reappeared on re-administration? | Not done | 0 |
| 5 | Alternative causes? | Unknown | 0 |
| 6 | Reappeared with placebo? | Unknown | 0 |
| 7 | Drug level in toxic range? | Unknown | 0 |
| 8 | Dose-response relationship? | Unknown | 0 |
| 9 | Similar previous reaction? | Unknown | 0 |
| 10 | Objective evidence? | Unknown | 0 |

**Total score: 3  Causality: Possible**

**Justification:** The adverse event occurred after administration of Ciprofloxacin, so the temporal relationship supports a positive answer to Naranjo question 2 (+2). The event improved after withdrawal/dechallenge, supporting question 3 (+1). Rechallenge was not done, so question 4 was scored 0. The remaining questions were marked Unknown because the supplied case information does not provide sufficient evidence to score them. No assumption was made from the presence of concomitant medicines.

**Conclusion:** The final Naranjo score is in the Possible range. The timing of the event supports a possible relationship, but the available information is not sufficient to establish a stronger causal association.

# SYN-PV-022 – Sertraline / Dizziness

**Patient:** 43 years, Female  **Dose:** 5 mg once daily  **Route:** Oral
**Indication:** Infection  **Onset:** Day 21
**Dechallenge:** Negative  **Rechallenge:** Positive
**Concomitant medicines:** Paracetamol

| Q | Naranjo question | Answer | Score |
|---|---|---|---|
| 1 | Previous conclusive reports? | Unknown | 0 |
| 2 | Event after drug administered? | Yes | 2 |
| 3 | Improved on withdrawal/antagonist? | Negative | 0 |
| 4 | Reappeared on re-administration? | Positive | 2 |
| 5 | Alternative causes? | Unknown | 0 |
| 6 | Reappeared with placebo? | Unknown | 0 |
| 7 | Drug level in toxic range? | Unknown | 0 |
| 8 | Dose-response relationship? | Unknown | 0 |
| 9 | Similar previous reaction? | Unknown | 0 |
| 10 | Objective evidence? | Unknown | 0 |

**Total score: 4  Causality: Possible**

**Justification:** The adverse event occurred after administration of Sertraline, so the temporal relationship supports a positive answer to Naranjo question 2 (+2). The recorded dechallenge was negative, so no point was assigned for question 3. A positive rechallenge was reported and therefore added +2 for question 4. The remaining questions were marked Unknown because the supplied case information does not provide sufficient evidence to score them. No assumption was made from the presence of concomitant medicines.

**Conclusion:** The final Naranjo score is in the Possible range. The timing of the event supports a possible relationship, but the available information is not sufficient to establish a stronger causal association.

# SYN-PV-023 – Omeprazole / Headache

**Patient:** 26 years, Male  **Dose:** 500 mg twice daily  **Route:** Oral
**Indication:** Anxiety  **Onset:** Day 4
**Dechallenge:** Negative  **Rechallenge:** Negative
**Concomitant medicines:** None reported

| Q | Naranjo question | Answer | Score |
|---|---|---|---|
| 1 | Previous conclusive reports? | Unknown | 0 |
| 2 | Event after drug administered? | Yes | 2 |
| 3 | Improved on withdrawal/antagonist? | Negative | 0 |
| 4 | Reappeared on re-administration? | Negative | -1 |
| 5 | Alternative causes? | Unknown | 0 |
| 6 | Reappeared with placebo? | Unknown | 0 |
| 7 | Drug level in toxic range? | Unknown | 0 |
| 8 | Dose-response relationship? | Unknown | 0 |
| 9 | Similar previous reaction? | Unknown | 0 |
| 10 | Objective evidence? | Unknown | 0 |

**Total score: 1  Causality: Possible**

**Justification:** The adverse event occurred after administration of Omeprazole, so the temporal relationship supports a positive answer to Naranjo question 2 (+2). The recorded dechallenge was negative, so no point was assigned for question 3. A negative rechallenge was reported and therefore gave -1 for question 4. The remaining questions were marked Unknown because the supplied case information does not provide sufficient evidence to score them. No assumption was made from the presence of concomitant medicines.

**Conclusion:** The final Naranjo score is in the Possible range. The timing of the event supports a possible relationship, but the available information is not sufficient to establish a stronger causal association.

## SYN-PV-024 – Losartan / Fatigue

**Patient:** 79 years, Unknown  **Dose:** 500 mg twice daily  **Route:** Oral
**Indication:** Hypertension  **Onset:** Day 3
**Dechallenge:** Positive  **Rechallenge:** Positive
**Concomitant medicines:** Paracetamol

| Q | Naranjo question | Answer | Score |
|---|---|---|---|
| 1 | Previous conclusive reports? | Unknown | 0 |
| 2 | Event after drug administered? | Yes | 2 |
| 3 | Improved on withdrawal/antagonist? | Positive | 1 |
| 4 | Reappeared on re-administration? | Positive | 2 |
| 5 | Alternative causes? | Unknown | 0 |
| 6 | Reappeared with placebo? | Unknown | 0 |
| 7 | Drug level in toxic range? | Unknown | 0 |
| 8 | Dose-response relationship? | Unknown | 0 |
| 9 | Similar previous reaction? | Unknown | 0 |
| 10 | Objective evidence? | Unknown | 0 |

**Total score: 5  Causality: Probable**

**Justification:** The adverse event occurred after administration of Losartan, so the temporal relationship supports a positive answer to Naranjo question 2 (+2). The event improved after withdrawal/dechallenge, supporting question 3 (+1). A positive rechallenge was reported and therefore added +2 for question 4. The remaining questions were marked Unknown because the supplied case information does not provide sufficient evidence to score them. No assumption was made from the presence of concomitant medicines.

**Conclusion:** The final Naranjo score is in the Probable range. The temporal relationship together with the available dechallenge/rechallenge information provides reasonable support for a causal relationship.

# SYN-PV-025 – Amoxicillin / Rash

**Patient:** 61 years, Unknown  **Dose:** 500 mg twice daily  **Route:** Oral
**Indication:** Infection  **Onset:** Day 21
**Dechallenge:** Negative  **Rechallenge:** Positive
**Concomitant medicines:** Paracetamol

| Q | Naranjo question | Answer | Score |
|---|---|---|---|
| 1 | Previous conclusive reports? | Unknown | 0 |
| 2 | Event after drug administered? | Yes | 2 |
| 3 | Improved on withdrawal/antagonist? | Negative | 0 |
| 4 | Reappeared on re-administration? | Positive | 2 |
| 5 | Alternative causes? | Unknown | 0 |
| 6 | Reappeared with placebo? | Unknown | 0 |
| 7 | Drug level in toxic range? | Unknown | 0 |
| 8 | Dose-response relationship? | Unknown | 0 |
| 9 | Similar previous reaction? | Unknown | 0 |
| 10 | Objective evidence? | Unknown | 0 |

**Total score: 4  Causality: Possible**

**Justification:** The adverse event occurred after administration of Amoxicillin, so the temporal relationship supports a positive answer to Naranjo question 2 (+2). The recorded dechallenge was negative, so no point was assigned for question 3. A positive rechallenge was reported and therefore added +2 for question 4. The remaining questions were marked Unknown because the supplied case information does not provide sufficient evidence to score them. No assumption was made from the presence of concomitant medicines.

**Conclusion:** The final Naranjo score is in the Possible range. The timing of the event supports a possible relationship, but the available information is not sufficient to establish a stronger causal association.

## SYN-PV-026 – Atorvastatin / Myalgia

**Patient:** 48 years, Unknown  **Dose:** 20 mg once daily  **Route:** Oral
**Indication:** Hypertension  **Onset:** Day 12
**Dechallenge:** Unknown  **Rechallenge:** Positive
**Concomitant medicines:** Vitamin D

| Q | Naranjo question | Answer | Score |
|---|---|---|---|
| 1 | Previous conclusive reports? | Unknown | 0 |
| 2 | Event after drug administered? | Yes | 2 |
| 3 | Improved on withdrawal/antagonist? | Unknown | 0 |
| 4 | Reappeared on re-administration? | Positive | 2 |
| 5 | Alternative causes? | Unknown | 0 |
| 6 | Reappeared with placebo? | Unknown | 0 |
| 7 | Drug level in toxic range? | Unknown | 0 |
| 8 | Dose-response relationship? | Unknown | 0 |
| 9 | Similar previous reaction? | Unknown | 0 |
| 10 | Objective evidence? | Unknown | 0 |

**Total score: 4  Causality: Possible**

**Justification:** The adverse event occurred after administration of Atorvastatin, so the temporal relationship supports a positive answer to Naranjo question 2 (+2). Dechallenge information was not available, so question 3 was scored 0. A positive rechallenge was reported and therefore added +2 for question 4. The remaining questions were marked Unknown because the supplied case information does not provide sufficient evidence to score them. No assumption was made from the presence of concomitant medicines.

**Conclusion:** The final Naranjo score is in the Possible range. The timing of the event supports a possible relationship, but the available information is not sufficient to establish a stronger causal association.

## SYN-PV-027 – Metformin / Nausea

**Patient:** 24 years, Unknown  **Dose:** 20 mg once daily  **Route:** Oral
**Indication:** Hypertension  **Onset:** Day 14
**Dechallenge:** Unknown  **Rechallenge:** Positive
**Concomitant medicines:** Paracetamol

| Q | Naranjo question | Answer | Score |
|---|---|---|---|
| 1 | Previous conclusive reports? | Unknown | 0 |
| 2 | Event after drug administered? | Yes | 2 |
| 3 | Improved on withdrawal/antagonist? | Unknown | 0 |
| 4 | Reappeared on re-administration? | Positive | 2 |
| 5 | Alternative causes? | Unknown | 0 |
| 6 | Reappeared with placebo? | Unknown | 0 |
| 7 | Drug level in toxic range? | Unknown | 0 |
| 8 | Dose-response relationship? | Unknown | 0 |
| 9 | Similar previous reaction? | Unknown | 0 |
| 10 | Objective evidence? | Unknown | 0 |

**Total score: 4  Causality: Possible**

**Justification:** The adverse event occurred after administration of Metformin, so the temporal relationship supports a positive answer to Naranjo question 2 (+2). Dechallenge information was not available, so question 3 was scored 0. A positive rechallenge was reported and therefore added +2 for question 4. The remaining questions were marked Unknown because the supplied case information does not provide sufficient evidence to score them. No assumption was made from the presence of concomitant medicines.

**Conclusion:** The final Naranjo score is in the Possible range. The timing of the event supports a possible relationship, but the available information is not sufficient to establish a stronger causal association.

# SYN-PV-028 – Amlodipine / Peripheral oedema

**Patient:** 77 years, Unknown  **Dose:** 5 mg once daily  **Route:** Oral
**Indication:** Hypertension  **Onset:** Day 18
**Dechallenge:** Unknown  **Rechallenge:** Negative
**Concomitant medicines:** Vitamin D

| Q | Naranjo question | Answer | Score |
|---|---|---|---|
| 1 | Previous conclusive reports? | Unknown | 0 |
| 2 | Event after drug administered? | Yes | 2 |
| 3 | Improved on withdrawal/antagonist? | Unknown | 0 |
| 4 | Reappeared on re-administration? | Negative | -1 |
| 5 | Alternative causes? | Unknown | 0 |
| 6 | Reappeared with placebo? | Unknown | 0 |
| 7 | Drug level in toxic range? | Unknown | 0 |
| 8 | Dose-response relationship? | Unknown | 0 |
| 9 | Similar previous reaction? | Unknown | 0 |
| 10 | Objective evidence? | Unknown | 0 |

**Total score: 1  Causality: Possible**

**Justification:** The adverse event occurred after administration of Amlodipine, so the temporal relationship supports a positive answer to Naranjo question 2 (+2). Dechallenge information was not available, so question 3 was scored 0. A negative rechallenge was reported and therefore gave -1 for question 4. The remaining questions were marked Unknown because the supplied case information does not provide sufficient evidence to score them. No assumption was made from the presence of concomitant medicines.

**Conclusion:** The final Naranjo score is in the Possible range. The timing of the event supports a possible relationship, but the available information is not sufficient to establish a stronger causal association.

# SYN-PV-029 – Ciprofloxacin / Tendon pain

**Patient:** 35 years, Unknown  **Dose:** 5 mg once daily  **Route:** Oral
**Indication:** Diabetes  **Onset:** Day 13
**Dechallenge:** Unknown  **Rechallenge:** Positive
**Concomitant medicines:** Paracetamol

| Q | Naranjo question | Answer | Score |
|---|---|---|---|
| 1 | Previous conclusive reports? | Unknown | 0 |
| 2 | Event after drug administered? | Yes | 2 |
| 3 | Improved on withdrawal/antagonist? | Unknown | 0 |
| 4 | Reappeared on re-administration? | Positive | 2 |
| 5 | Alternative causes? | Unknown | 0 |
| 6 | Reappeared with placebo? | Unknown | 0 |
| 7 | Drug level in toxic range? | Unknown | 0 |
| 8 | Dose-response relationship? | Unknown | 0 |
| 9 | Similar previous reaction? | Unknown | 0 |
| 10 | Objective evidence? | Unknown | 0 |

**Total score: 4  Causality: Possible**

**Justification:** The adverse event occurred after administration of Ciprofloxacin, so the temporal relationship supports a positive answer to Naranjo question 2 (+2). Dechallenge information was not available, so question 3 was scored 0. A positive rechallenge was reported and therefore added +2 for question 4. The remaining questions were marked Unknown because the supplied case information does not provide sufficient evidence to score them. No assumption was made from the presence of concomitant medicines.

**Conclusion:** The final Naranjo score is in the Possible range. The timing of the event supports a possible relationship, but the available information is not sufficient to establish a stronger causal association.

## SYN-PV-030 – Sertraline / Dizziness

**Patient:** 82 years, Male  **Dose:** 20 mg once daily  **Route:** Oral
**Indication:** Anxiety  **Onset:** Day 17
**Dechallenge:** Positive  **Rechallenge:** Positive
**Concomitant medicines:** Paracetamol

| Q | Naranjo question | Answer | Score |
|---|---|---|---|
| 1 | Previous conclusive reports? | Unknown | 0 |
| 2 | Event after drug administered? | Yes | 2 |
| 3 | Improved on withdrawal/antagonist? | Positive | 1 |
| 4 | Reappeared on re-administration? | Positive | 2 |
| 5 | Alternative causes? | Unknown | 0 |
| 6 | Reappeared with placebo? | Unknown | 0 |
| 7 | Drug level in toxic range? | Unknown | 0 |
| 8 | Dose-response relationship? | Unknown | 0 |
| 9 | Similar previous reaction? | Unknown | 0 |
| 10 | Objective evidence? | Unknown | 0 |

**Total score: 5  Causality: Probable**

**Justification:** The adverse event occurred after administration of Sertraline, so the temporal relationship supports a positive answer to Naranjo question 2 (+2). The event improved after withdrawal/dechallenge, supporting question 3 (+1). A positive rechallenge was reported and therefore added +2 for question 4. The remaining questions were marked Unknown because the supplied case information does not provide sufficient evidence to score them. No assumption was made from the presence of concomitant medicines.

**Conclusion:** The final Naranjo score is in the Probable range. The temporal relationship together with the available dechallenge/rechallenge information provides reasonable support for a causal relationship.

## Assessment Note

This report is based only on the fictional cases and information supplied in the Workora case pack. Unknown information has not been filled in by assumption. The causality category may change if additional clinical or follow-up information becomes available.